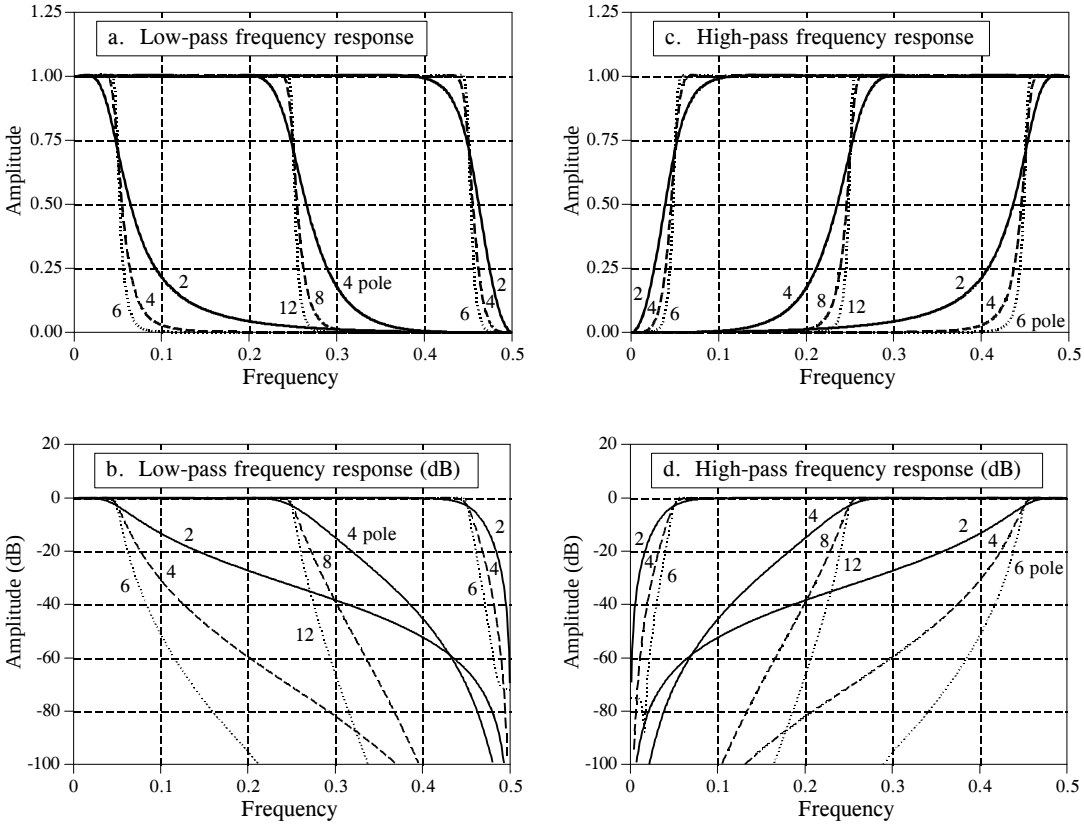


FIGURE 20-2
Chebyshev frequency responses. Figures (a) and (b) show the frequency responses of low-pass Chebyshev filters with 0.5% ripple, while (c) and (d) show the corresponding high-pass filter responses.

Kidding aside, the point is that you can use these filters very effectively without knowing the nasty mathematics behind them. Filter design is a specialty. In actual practice, more engineers, scientists and programmers think in terms of answer 2, than answer 1.

Figure 20-2 shows the frequency response of several Chebyshev filters with 0.5% ripple. For the method used here, the number of poles must be *even*. The cutoff frequency of each filter is measured where the amplitude crosses 0.707 (-3dB). Filters with a cutoff frequency near 0 or 0.5 have a sharper roll-off than filters in the center of the frequency range. For example, a two pole filter at $f_C = 0.05$ has about the same roll-off as a four pole filter at $f_C = 0.25$. This is fortunate; fewer poles can be used near 0 and 0.5 because of round-off noise. More about this later.

There are two ways of finding the recursion coefficients without using the z-transform. First, the cowards way: use a table. Tables 20-1 and 20-2 provide the recursion coefficients for low-pass and high-pass filters with 0.5% passband ripple. If you only need a quick and dirty design, copy the appropriate coefficients into your program, and you're done.